Experiment No. 1 : Implementation of Data Preprocessing using Python

```
import pandas as pd
import matplotlib.pylab as plt
import numpy as np
```

Load the dataset using pandas library

```
df = pd.read_csv("autodata.csv")
```

Check the contents of dataset using df.head() and df.tail() functions

```
df.head(5)
```

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels
0	0	3	122	alfa-romero	std	two	convertible	rwd
1	1	3	122	alfa-romero	std	two	convertible	rwd
2	2	1	122	alfa-romero	std	two	hatchback	rwd
3	3	2	164	audi	std	four	sedan	fwd
4	4	2	164	audi	std	four	sedan	4wd

5 rows × 30 columns

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```
df.tail(5)
```

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	l
196	196	-1	95	volvo	std	four	sedan	rwd	
197	197	-1	95	volvo	turbo	four	sedan	rwd	
198	198	-1	95	volvo	std	four	sedan	rwd	
199	199	-1	95	volvo	turbo	four	sedan	rwd	
200	200	-1	95	volvo	turbo	four	sedan	rwd	

5 rows × 30 columns

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```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 201 entries, 0 to 200
```

```
Data columns (total 30 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         201 non-null    int64
 1   symboling          201 non-null    int64
 2   normalized-losses  201 non-null    int64
 3   make               201 non-null    object
 4   aspiration         201 non-null    object
 5   num-of-doors       201 non-null    object
 6   body-style         201 non-null    object
 7   drive-wheels       201 non-null    object
 8   engine-location    201 non-null    object
 9   wheel-base         201 non-null    float64
 10  length             201 non-null    float64
 11  width              201 non-null    float64
 12  height             201 non-null    float64
 13  curb-weight        201 non-null    int64
 14  engine-type        201 non-null    object
 15  num-of-cylinders   201 non-null    object
 16  engine-size        201 non-null    int64
 17  fuel-system        201 non-null    object
 18  bore               201 non-null    float64
 19  stroke             197 non-null    float64
 20  compression-ratio  201 non-null    float64
 21  horsepower         199 non-null    float64
 22  peak-rpm           199 non-null    float64
 23  city-mpg           201 non-null    int64
 24  highway-mpg        201 non-null    int64
 25  price              201 non-null    int64
 26  city-L/100km       201 non-null    float64
 27  horsepower-binned  199 non-null    object
 28  diesel             201 non-null    int64
 29  gas                201 non-null    int64
dtypes: float64(10), int64(10), object(10)
memory usage: 47.2+ KB
```

.

```
df.describe()
```

	Unnamed: 0	symboling	normalized-losses	wheel-base	length	width	
count	201.000000	201.000000	201.00000	201.000000	201.000000	201.000000	20
mean	100.000000	0.840796	122.00000	98.797015	0.837102	0.915126	5
std	58.167861	1.254802	31.99625	6.066366	0.059213	0.029187	
min	0.000000	-2.000000	65.00000	86.600000	0.678039	0.837500	4
25%	50.000000	0.000000	101.00000	94.500000	0.801538	0.890278	5
50%	100.000000	1.000000	122.00000	97.000000	0.832292	0.909722	5
75%	150.000000	2.000000	137.00000	102.400000	0.881788	0.925000	5
max	200.000000	3.000000	256.00000	120.900000	1.000000	1.000000	5

Evaluating for Missing Data

```
df.isnull()
```

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	l
0	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	
196	False	False	False	False	False	False	False	False	
197	False	False	False	False	False	False	False	False	
198	False	False	False	False	False	False	False	False	
199	False	False	False	False	False	False	False	False	
200	False	False	False	False	False	False	False	False	

201 rows × 30 columns

.

```
df.notnull()
```

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	l
0	True	True	True	True	True	True	True	True	
1	True	True	True	True	True	True	True	True	
2	True	True	True	True	True	True	True	True	
3	True	True	True	True	True	True	True	True	
4	True	True	True	True	True	True	True	True	
...	...	...	...	...	...	...	...	...	
196	True	True	True	True	True	True	True	True	
197	True	True	True	True	True	True	True	True	
198	True	True	True	True	True	True	True	True	
199	True	True	True	True	True	True	True	True	
200	True	True	True	True	True	True	True	True	

201 rows × 30 columns

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```
df.notnull().sum()
```

	0
Unnamed: 0	201
symboling	201
normalized-losses	201
make	201
aspiration	201
num-of-doors	201
body-style	201
drive-wheels	201
engine-location	201
wheel-base	201
length	201
width	201
height	201
curb-weight	201
engine-type	201
num-of-cylinders	201
engine-size	201
fuel-system	201
bore	201
stroke	197
compression-ratio	201
horsepower	199
peak-rpm	199
city-mpg	201
highway-mpg	201
price	201
city-L/100km	201
horsepower-binned	199
diesel	201
gas	201

dtype: int64

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